BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2025

BIOLOGY

Past Paper 2025

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Biology (Objective Type) - Group - I

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The destruction of the walls of alveoli is:

(A) Bronchitis
(B) Emphysema
(C) Pneumonia
(D) Asthma

Q2. Treatment for the end-stage kidney failure is:

(A) Lithotripsy
(B) Peritoneal dialysis
(C) Haemodialysis
(D) Kidney transplant

Q3. Hormone responsible for male secondary sex characters is:

(A) Estrogen
(B) Progesterone
(C) Testosterone
(D) Glucagon

Q4. The inflammation of the membranes at joints is:

(A) Osteoporosis
(B) Osteoarthritis
(C) Rheumatoid arthritis
(D) Gout

Q5. Vegetative propagation by leaves is found in:

(A) Bryophyllum
(B) Mint
(C) Yams
(D) Calocacia

Q6. An insect pollinated flower is:

(A) Grass
(B) Sunflower
(C) Willow
(D) Corn

Q7. The genotype having identical alleles in a gene pair is called:

(A) Azygous
(B) Hemizygous
(C) Heterozygous
(D) Homozygous

Q8. Theory of natural selection was proposed by:

(A) Charles Darwin
(B) C de Buffon
(C) J. de Lamarck
(D) Gregor Mendel

Q9. The self-sufficient unit of an environment is:

(A) Biosphere
(B) Ecosystem
(C) Community
(D) Population

Q10. The principal building block of many biomolecules is:

(A) Nitrogen
(B) Hydrogen
(C) Oxygen
(D) Carbon

Q11. In glycolysis the glucose molecule is broken into two molecules of:

(A) Ethanol
(B) Lactic acid
(C) Pyruvic acid
(D) Acetic acid

Q12. Mescaline comes from:

(A) Mushroom
(B) Cactus
(C) Cannabis
(D) Morning glory

Biology (Subjective) (PART - I) - Group - I

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Write symptoms of pneumonia.

(ii) How smoking effects badly on human health? Write any two effects.

(iii) How will you differentiate between acute bronchitis and chronic bronchitis?

(iv) Define cellular respiration.

(v) What is lithotripsy?

(vi) Write any two major causes of kidney failure.

(vii) Write down functions of hypothalamus.

(viii) Write name of three parts of inner ear of man.

Q3. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Write name of ends of skeletal muscle.

(ii) What are the characteristics of gout?

(iii) Explain corms.

(iv) Write disadvantages of vegetative propagation in plants.

(v) What is co-dominance? Give example.

(vi) What is meant by inheritance in biology?

(vii) Define homologous chromosomes.

(viii) What is meant by nucleosomes?

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Describe two methods of nitrogen fixation.

(ii) Compare mutualism and commensalism.

(iii) Compare ecosystem and biosphere.

(iv) Write two effects of global warming.

(v) Write uses of glycerol.

(vi) How gene of interest is isolated in genetic engineering?

(vii) What are antibiotics? Also give example.

(viii) Define pharmacology and pharmacologist.

Biology (Subjective) (PART - II) - Group - I

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

> Note: Attempt any TWO questions.

Q5(a). (Marks 5) Write detail of kidney transplant.

Q5(b). (Marks 4) Explain components of human skeleton.

Q6(a). (Marks 5) Write a note on nerve cell or neuron.

Q6(b). (Marks 4) Discuss the techniques of tissue culture and cloning.

Q7(a). (Marks 5) Describe the achievements of genetic engineering in medicine, agriculture and environment.

Q7(b). (Marks 4) Write a note on resistance against antibiotics.

GROUP – II

Biology (Objective Type) - Group - II

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Stomata are present in:

(A) Xylem
(B) Phloem
(C) Epidermis
(D) Endodermis

Q2. Length of human kidney is:

(A) 10 cm
(B) 5 cm
(C) 4 cm
(D) 27 cm

Q3. The unit of nervous system is:

(A) Nucleus
(B) Neuron
(C) Receptors
(D) Nephron

Q4. Number of soft bones in new born babies:

(A) 200
(B) 206
(C) 256
(D) 300

Q5. Root is made from:

(A) Testa
(B) Plumule
(C) Micropyle
(D) Radical

Q6. Example of secondary consumers is:

(A) Livestock
(B) Snake
(C) Goat
(D) Cow

Q7. Example of corms is:

(A) Ginger
(B) Potato
(C) Onion
(D) Garlic

Q8. Number of hydrogen bonds in adenine and thyamine:

(A) Two (B) Three

(C) Four (D) One

Q9. Example of codominance is:

(A) Blood groups (B) Height

(C) Weight (D) Intelligence

Q10. Carbonates in crust of soil also form:

(A) O_2 (B) CO_2

(C) H_2O (D) H_2

Q11. Dolly is name of:

(A) Rat (B) Sheep

(C) Flower (D) Rabbit

Q12. When does E-Coli bacteria is created:

(A) 1980 (B) 1975

(C) 1977 (D) 1970

Biology (Subjective) (PART - I) - Group - II

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) How lungs are protected from free expanding and contracting movements?

(ii) What is asthma? Give its symptoms.

(iii) Write two bad effects of smoking.

(iv) Differentiate acute bronchitis and chronic bronchitis.

(v) What are succulent organs? Give an example.

(vi) How kidneys do pressure filtration?

(vii) What is the reason of night blindness?

(viii) Why adrenaline is termed as "emergency hormone"?

Q3. (Marks 10) Write short answers to any FIVE (5) questions:

(i) What are tendons? Write their function.

(ii) Write two examples of hinge joints.

(iii) Define epigeal germination.

(iv) Write function of Cowper's glands.

(v) Describe phenotype ratio of F2 generation derived by Mendel's Law of Segregation.

(vi) Why Mendel selected pea plants for his experiments?

(vii) Write two sources of variations.

(viii) Define artificial selection.

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

(i) If producers are eliminated from ecosystem, what will happen?

(ii) Define biosphere.

(iii) Briefly describe global warming.

(iv) How biogeochemical cycles play their role in maintaining an ecosystem?

(v) What is meant by fermentation?

(vi) Define glycolysis.

(vii) What is the function of antibodies in blood?

(viii) Define sedatives and give one example.

Biology (Subjective) (PART - II) - Group - II

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

> Note: Attempt any TWO questions.

Q5(a). (Marks 5) Write a note on kidney stone formation and its treatment.

Q5(b). (Marks 4) Write detail of arthritis.

Q6(a). (Marks 5) Explain autonomic nervous system.

Q6(b). (Marks 4) Describe the structure of a typical dicot seed.

Q7(a). (Marks 5) Write basic steps in genetic engineering.

Q7(b). (Marks 4) Write a note on narcotics and their effects.